

GCF-NSD Lab Request Package v0.1

Safe External Version / No Internal Inference Framework Disclosed

Purpose

This document is intended only for technical feasibility discussion and preliminary laboratory quotation. It is not a final alloy composition, production specification, certified performance claim, or disclosure of any proprietary scoring, inference, or cross-domain design framework.

1. External Positioning

This is an early-stage feasibility request. The goal is to evaluate whether a weldable HSLA-like marine/offshore structural steel matrix can support coupon-level samples that improve corrosion-fatigue decoupling and crack-arrest behavior while preserving weld / HAZ stability.

The first round is not intended to produce a final commercial alloy composition, production specification, or naval/military performance claim. The goal is to identify whether candidate routes produce a measurable feasibility signal relative to a commercial marine/offshore baseline.

2. One-Page Summary

- Project title: Coupon-Level Feasibility Study for Coupled Failure Suppression in HSLA-like Marine/Offshore Structural Steel.
- Target failure chain: localized corrosion / pitting -> fatigue crack initiation -> crack connectivity / growth -> weld or HAZ-related amplification.
- Primary concept: compare CA, CF, and CA-CF functional directions on an HSLA-like matrix.
- CA: crack-arrest, crack-deflection, crack-blunting, or increased crack-path tortuosity.
- CF: reduced pitting-to-fatigue transition or reduced corrosion-fatigue coupling.

3. Baseline Requirement

The baseline should not be selected merely for convenience or to maximize apparent improvement. A dual-baseline logic is preferred:

- B0-A: an experimentally available commercial marine/offshore structural steel, preferably API 2W Grade 50 or API 2H Grade 50. If unavailable, ASTM A131 / ABS DH36 or EH36 may be considered.
- B0-B: a higher-level literature or strategic reference, such as HSLA-80 / HSLA-100-class naval structural steel. B0-B does not need to be physically tested in the first round, but it should constrain interpretation.
- Any improvement over B0-A should be reported only as a feasibility signal, not as proof of superiority over high-end naval steel benchmarks.

4. Suggested Sample Groups and Coupon Conditions

- B0-A: commercial marine/offshore baseline.
- S1 CA-biased candidate: a route intended to improve crack arrest, crack deflection, crack blunting, or crack-path tortuosity.
- S2 CF-biased candidate: a route intended to reduce pitting-to-fatigue transition or corrosion-fatigue coupling.

- S3 CA-CF candidate: a route intended to combine crack-arrest behavior and corrosion-fatigue decoupling behavior.
- S4 optional: a HAZ/process-adjusted CA-CF route if the laboratory expects S3 to create weld or HAZ instability.

For each material group, at minimum, base metal coupons and simulated HAZ coupons or welded-region proxy coupons are requested. The purpose is to ensure that any positive signal in base metal does not collapse under HAZ-like conditions.

Group	Base Metal Coupon	HAZ-like / Weld Proxy Coupon	Purpose
B0-A baseline	Required	Required	Commercial marine/offshore steel control
S1 CA-biased	Required	Required	Crack-arrest / deflection / blunting direction
S2 CF-biased	Required	Required	Corrosion-fatigue decoupling direction
S3 CA-CF	Required	Required	Main candidate: combined CA + CF benefit
S4 optional	Optional	Required if included	HAZ/process-adjusted CA-CF route

5. Minimum Test Menu

Tier 1: Low-Cost Screening

- microstructure observation
- hardness mapping
- basic tensile properties
- impact or toughness proxy if feasible
- corrosion exposure
- pit density / pit depth observation
- HAZ-like hardness gradient

If a candidate route shows obvious embrittlement, excessive hardness gradient, unstable HAZ behavior, or severe process sensitivity at Tier 1, it should not proceed to Tier 2.

Tier 2: Core Feasibility Tests

- Corrosion-fatigue proxy: fatigue after corrosion pre-exposure, crack initiation cycle comparison, and percentage of cracks initiating from pits.
- Crack-arrest proxy: crack path tortuosity, crack deflection or branching, crack growth rate proxy, and fracture surface observation.
- HAZ stability proxy: the same comparison under simulated HAZ or welded-region proxy conditions, HAZ hardness gradient, and retention of fatigue / corrosion-fatigue behavior after HAZ-like treatment.

6. Testing Priority

1. B0-A versus S3 CA-CF under corrosion pre-exposure plus fatigue or crack-initiation proxy.
2. B0-A versus S3 CA-CF under HAZ-like condition.
3. S1 versus S2 versus S3 crack-path morphology comparison.
4. Expanded repeat coupons or more rigorous fatigue crack growth testing.

7. Interpretation Rules

Positive Feasibility Signal

- Improved or non-degraded fatigue behavior after corrosion pre-exposure relative to B0-A.
- Reduced or delayed pit-origin crack initiation.
- Increased crack path tortuosity, deflection, branching, or reduced connectivity.
- No major degradation under HAZ-like condition.
- Improvement not explainable solely by increased hardness or strength.

Boundary Signal

- Static corrosion rate improves, but fatigue after corrosion does not improve.
- Strength improves, but crack path becomes straighter or HAZ behavior worsens.
- Base metal improves, but HAZ-like coupon degrades.
- S3 performs no better than S1 or S2.
- Only one isolated coupon shows improvement.

Negative / Stop Signal

- S3 corrosion-fatigue performance is worse than B0-A.
- HAZ-like coupon degrades significantly.
- Crack path becomes straighter, faster, or more connected.
- All improvement can be explained by hardness / strength increase.
- The laboratory cannot provide clear baseline, coupon conditions, environment, loading history, or repeatability information.

8. Requested Laboratory Feedback

5. Which commercial steel grade would you recommend as B0-A, and why?
6. Can API 2W Grade 50, API 2H Grade 50, ABS DH36, or ABS EH36 be sourced or approximated?
7. What feasible microstructure or processing routes could produce CA-biased behavior?
8. What feasible microstructure, surface, interface, or processing routes could produce CF-biased behavior?
9. Is a CA-CF combined route technically plausible without harming weldability or HAZ stability?
10. Can simulated HAZ coupons or welded-region proxy coupons be prepared?
11. What is the minimum practical test package for observing corrosion-fatigue and crack-path behavior?
12. What would be the approximate cost and time for Tier 1 only?
13. What would be the approximate cost and time for Tier 1 plus limited Tier 2?
14. What results would be sufficient to justify a second-round expanded study?
15. What risks would make this concept impractical?
16. What information would be needed before preparing a formal quotation?

9. Confidentiality and Disclosure Boundary

Disclosure Boundary

This document is intended only for technical feasibility discussion. It does not disclose any final alloy composition, heat-treatment recipe, proprietary scoring method, internal inference framework, commercial deployment plan, or certified naval/military performance claim. Before exchanging detailed candidate routes, experimental data, or patent-relevant implementation details, a confidentiality agreement is recommended.

Suggested use: provide the safe lab brief v0.2 for initial contact. Provide this Lab Request Package only after the recipient is willing to evaluate feasibility, cost, and technical route options.